

strataq: measurement instruments for stochastic strategic systems

Susceptibility, reciprocity, and entropy production for quantal-response equilibria, calibrated on games with known answers

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Abstract

We present **strataq**, an open JAX library that treats logit quantal-response equilibrium (QRE) analysis as *instrumentation*: a susceptibility meter, a reciprocity meter, an entropy-production meter, and a phase locator, each an executable function with a stated confidence tier and a regenerable calibration artifact. The exact backbone is known (QRE = entropy-regularised response; Gibbs stationarity in potential games); the library’s contribution is (i) the *reciprocity transfer*: the observable equilibrium response $\chi^{\text{eq}} = (I - SB)^{-1}S$ is symmetric *iff* the normalised game has zero harmonic component, so strategic feedback neither creates nor destroys reciprocity—making the reciprocity defect $\mathcal{R}$ a payoff-free, λ -free *test* of potentiality; (ii) a separable Kronecker implementation of the Candogan decomposition giving the harmonic fraction α in near-linear time; (iii) exact-tier dynamics meters (currents, entropy production) on the joint-profile Glauber chain; and (iv) calibration of every meter at both ends of the α axis with payoffs known by construction: Rosenthal congestion on real Sioux Falls network data ($\mathcal{R} = 5.7 \times 10^{-17}$, Fisk KKT 7×10^{-15}) and Colonel Blotto ($\alpha = 0.69$, $\mathcal{R} = 0.12$). Pointing the calibrated instruments at seeded game families produced three findings we did not predict: $\mathcal{R}$ magnitudes scale with λ (only the zero test is λ -free); potential games *escape* criticality at high λ through equilibrium concentration, opening a supercritical wedge only for $\alpha \gtrsim 0.5$; and the response and dissipation layers decouple structurally in the near-harmonic regime—the correlation between response asymmetry and entropy production reverses sign at $\alpha = 0.95$, and no renormalisation of the response matrix repairs it. Every quantitative statement in this paper regenerates from fixed seeds (**make reproduce**); Appendix A maps each number to its artifact.

1 Introduction

Quantal response equilibrium [1] replaced exact best response with logit response, and in doing so imported the entire apparatus of Gibbs statistical mechanics into strategic analysis: the logit choice rule is the Gibbs variational principle, its log-partition function is a cumulant generating function, and in potential games [4] the induced Glauber dynamics are reversible with stationary measure $\pi \propto e^{\lambda\Phi}$ [5]. What the QRE literature has largely not done is treat this correspondence *instrumentally* — as licensing a set of measuring devices whose readings are meaningful whether or not they match a prediction.

*With agent-assisted implementation. Code, gate records, and every number’s regeneration path: <https://github.com/SharathSPhD/sage>.

This paper describes such a set of devices and what happened when we pointed them at things. The stance is deliberate: this is not a hypothesis-testing exercise. We built a susceptibility meter, a reciprocity meter, an entropy-production meter, and a phase locator; we calibrated each on systems where the correct reading is known analytically; and then we swept them across a one-parameter family of strategic environments — the harmonic fraction α of the Candogan decomposition [6] — and wrote down what they read. Three of the readings surprised us, and one of them refuted a correction we had proposed to our own instrument. All of it is reported, including the refutation.

Two disciplines make this more than a metaphor. First, every claim carries a confidence tier — *exact* (proved identity, implemented and verified to 10^{-12} – 10^{-16}), *derived* (proved in-house, numerically checked), *conjectured* (with a stated falsifier), or *speculative* — and the tier is visible in the code’s docstrings, the project’s claim ledger, and this paper. Second, every number below regenerates from a fixed seed by one command; Appendix A maps each to its machine-readable artifact, and each artifact was closed under an adversarial review whose objections and dispositions are part of the public record.

The library is **strataq**; the umbrella project is SAGE (Strategic Agent Game Engine). The research programme it instruments — non-equilibrium thermodynamics of strategic systems, with empirical estimation across pricing, energy, congestion and allocation domains — is described in the repository’s research documents; this paper is the toolkit paper, written first because the toolkit is, we believe, the most immediately useful artifact.

2 The instruments and their tiers

Notation: players $i \in \{1, \dots, N\}$ with finite action sets, mixed profiles σ_i , logit precision λ_i ; $C_i = \text{diag}(\sigma_i) - \sigma_i \sigma_i^\top$; $S = \text{blockdiag}(\lambda_i C_i)$; $B_{ij}(a, b) = \partial U_i(a) / \partial \sigma_j(b)$ the cross-payoff operator. All linear algebra involving $(I - SB)$ is performed on the tangent space $T = \bigoplus_i \{v : \mathbf{1}^\top v = 0\}$ via an explicit Helmert basis; C_i is rank-deficient by construction, and skipping this projection manufactures a spurious zero eigenvalue — a false criticality reading. Decomposition always acts on the *normalised* (effective) game: full externality symmetry is sufficient but not necessary for potentiality of the raw game [9].

2.1 The susceptibility meter (exact + derived)

The opponents-frozen response to an own-payoff field h_i is the static fluctuation–dissipation identity $\chi^{\text{part}} = \lambda_i C_i$ (exact; verified to 10^{-12} by autodiff of the log-partition). With strategic feedback, total differentiation of the fixed point gives the *strategic resolvent*

$$\chi^{\text{eq}} = \frac{d\sigma^*}{dh} = (I - SB)^{-1} S \quad \text{on } T, \quad (1)$$

likely folklore as comparative statics (we cite defensively) but load-bearing here: the same operator is the contraction certificate ($\rho(SB) < 1 \Rightarrow$ unique QRE), the bifurcation detector (eigenvalues of SB crossing 1), and the implicit-differentiation kernel (our custom VJP solves the transposed tangent system; its Jacobian reproduces (1) to machine epsilon). Verification: χ^{eq} matches central finite differences of the re-solved QRE to 1.3×10^{-8} over 50 random games.

2.2 The reciprocity meter (derived)

Result 1 (Reciprocity transfer). χ^{eq} is symmetric $\iff S(B - B^\top)S = 0 \iff$ (full support) $B = B^\top$ on $T \iff$ the normalised game has zero harmonic component.

The proof is three lines of resolvent algebra; the *content* is that strategic feedback neither creates nor destroys reciprocity, so the observable equilibrium response inherits the symmetry of the unobservable payoff operator exactly. A prior-art sweep (recorded in the repository, including the search scope) found no prior statement. Define the reciprocity defect $\mathcal{R} = \|\chi^{\text{eq}} - \chi^{\text{eq}\top}\|_F / \|\chi^{\text{eq}} + \chi^{\text{eq}\top}\|_F \in [0, \infty)$. Because Result 1 is a symmetry statement, *whether* $\mathcal{R} = 0$ is λ -free — nothing for a noise parameter to absorb, which is our lead response to the Haile et al. [3] non-falsifiability critique. Two properties discovered in calibration and now documented on the instrument: $\mathcal{R}$ is a norm ratio, unbounded above (matching pennies reads 1.2); and its *magnitude* scales with λ (approximately $\propto \lambda$ at small λ), so only the zero test is λ -robust.

2.3 The harmonic-fraction meter (exact)

The game graph is a Cartesian product of complete graphs; its Laplacian is a Kronecker sum with closed-form eigenspaces indexed by which coordinates vary. The Candogan projections therefore reduce to separable per-axis centering/averaging — an FFT-like transform, $O(\text{tensor size} \times N)$, no iterative solver — with the potential/harmonic split of each subset sector given by the degree-weighted mean across the players in the sector. We verify orthogonality (in the response-graph inner product), idempotence, exact reconstruction, and invariance of α under strategic equivalence, each to 10^{-9} or better. Families at exact target α are generated by mixing unit-norm potential and harmonic components — never by perturbing a Jacobian, whose skew part is coordinate-dependent [9].

2.4 The dissipation meter (exact)

Single-site logit revision on the joint profile space is a Glauber chain whose stationary distribution, probability currents J^* , and entropy production rate σ_{EP} we compute exactly (dense path, small games). In an exact potential game the revision probabilities are heat-bath conditionals of $\pi \propto e^{\lambda\Phi}$: detailed balance holds transition-by-transition (verified to 1.5×10^{-16}), $J^* = 0$, $\sigma_{\text{EP}} = 0$. Off potentiality the chain holds a non-equilibrium steady state that circulates and dissipates.

2.5 Solvers and the phase locator

Damped iteration, magnetic mirror descent [13] (last-iterate; agrees with damped to 7×10^{-13}), and pseudo-arclength continuation of the principal branch (agrees with Gambit’s homotopy [12] to 10^{-8} , and passes through the coordination pitchfork without stalling). The spectrum of SB on T supplies ρ , the distance to criticality $1 - \rho$, and bifurcation typing: $B = B^\top$ forces a real spectrum, so sustained cycles under logit dynamics cannot arise in full potential games — verified: the spectrum is exactly real on every potential game tested across $\lambda \in [0.5, 10]$, and visibly complex on harmonic games (cf. the logit limit cycles of Hommes and Ochea [15] at $\alpha = 1$).

3 Calibration: games with known answers

A meter is only as credible as the systems on which its reading is known. The α axis offers exactly two families with payoffs *free by construction* — no demand estimation contaminating the reading — and we anchored both ends before pointing the instruments anywhere else.

3.1 The $\alpha = 0$ anchor: congestion on real network data

Rosenthal congestion games [11] are exact potential games; in the population form, logit route choice *is* Fisk’s stochastic user equilibrium [10], whose convex potential (Beckmann integral plus scaled entropy) is known analytically. We implement the population engine (payoff field $F(x)$, potentiality tested by symmetry of DF — Sandholm’s externality symmetry [7], cited, not claimed) and solve SUE by damped Newton on the convex program in per-OD tangent coordinates; the damped logit-assignment map itself is *not* a contraction on steep-BPR networks (it two-cycles on the Braess diamond), a practical fact we document rather than hide.

Calibration readings (Table 1): on the Braess diamond, the autodiff gradient of the Beckmann integral equals the route-cost vector to 10^{-12} (an identity, and our sharpest end-to-end code check); on *real Sioux Falls data* (76 links, top-8 OD pairs, $k=3$ route sets from the TNTTP repository [17]), the Fisk KKT conditions hold to 7×10^{-15} , DF is symmetric to machine zero, and the toll-response reciprocity defect reads $\mathcal{R} = 5.7 \times 10^{-17}$. Link tolls enter route costs exactly linearly — the cleanest conjugate field in the programme — and the toll susceptibility matches finite differences to 2.8×10^{-9} , with the sign verified against actual re-solves. We state the scope once: route sets are k -shortest restrictions and the OD set is a subset, so computed link flows are *not* comparable to the repository’s best-known equilibrium flows (a diagnostic gap of 0.73 is reported as a diagnostic, never a certificate); the instrument identities are exact within the route set regardless.

3.2 The $\alpha > 0$ anchor: Colonel Blotto

Two-player Blotto over lattice-simplex allocation grids, budgets set by the experimenter (an exactly observable payoff perturbation). Readings: symmetric budget-3, three equal fields — $\alpha = 0.694$, $\mathcal{R} = 0.118$ at $\lambda = 1.5$; asymmetric field values — $\sigma_{EP} = 0.098$ with visible circulation. A degenerate instance (equal values, budget 2) has *constant* payoffs — every profile ties — and every meter reads exactly zero ($\sigma_{EP} = 5.4 \times 10^{-32}$): a correct null found while writing the positive test, kept as a regression. One calibration surprise worth naming: Blotto is *mixed*, not pure harmonic ($\alpha \approx 0.7$, not 1) — zero-sum does not imply harmonic-pure; RPS ($\alpha = 1$ exactly) remains the pure extreme.

4 Findings from pointing the instruments

With both anchors in place we swept seeded two-player 3×3 families of exact target α (mixed from unit-norm potential and harmonic components) and read every meter. Three findings follow, in the order discovered. All are rank-order statements at fixed $\lambda = 1.2$ on this family class unless noted; per-cell significance is neither claimed nor needed — surfaces and within-level correlations carry the content, with bootstrap intervals in the artifacts.

Reading	Requirement	Measured	Artifact
$\nabla\psi = \lambda\sigma, \nabla^2\psi = \lambda^2C$	10^{-12}	holds	test suite
Gibbs $\pi \propto e^{\lambda\Phi}$ (congestion, $N=2,3$)	10^{-10}	2.5×10^{-16}	<code>gibbs_agreement</code>
$\sigma_{EP}, J^* $ on potential games	$< 10^{-12}$	holds	<code>equilibrium_reads_zero</code>
$\mathcal{R}$ on 5 exact potential games	$< 10^{-10}$	$\leq 9 \times 10^{-17}$	<code>reciprocity_potential</code>
$\mathcal{R}$ on RPS-3/RPS-5/matching pennies	> 0.1	0.69/0.43/1.20	<code>reciprocity_harmonic</code>
χ^{eq} vs finite differences (50 games)	10^{-6}	1.3×10^{-8}	<code>chi_fd_agreement</code>
Real spectrum of SB , potential games, $\lambda \leq 10$	rel. imag $< 10^{-8}$	0	<code>spectrum_reality</code>
Gambit agreement, 32 games $\times 2 \lambda$	10^{-8}	1.2×10^{-9}	<code>gambit_agreement</code>
Sioux Falls: $\mathcal{R} / DF$ defect / KKT	$10^{-10}/10^{-12}/10^{-8}$	$5.7 \times 10^{-17}/0/7 \times 10^{-15}$	<code>sioux_falls_calibration</code>
Blotto: $\alpha, \mathcal{R}$, degenerate null	see text	0.69, 0.12, 5×10^{-32}	<code>blotto_readings</code>

Table 1: Calibration summary. Every row regenerates from seed 20260808 via `make reproduce`; artifacts are JSON records under `benchmarks/results/` (Appendix A).

4.1 Finding 1: co-movement along α , with a marginal-conditional trap

Marginally, the meters co-move almost perfectly: Spearman $\rho(\mathcal{R}, \alpha) = 0.982$ ($n=2000$), $\rho(\sigma_{EP}, \alpha) = 0.990$ and $\rho(\sigma_{EP}, \mathcal{R}) = 0.993$ ($n=1000$). The last number is a trap: stratified by α , the within-level coupling $\rho(\sigma_{EP}, \mathcal{R} \mid \alpha)$ is $+0.80$ to $+0.88$ for $\alpha \leq 0.65$, degrades through $+0.61$ ($\alpha=0.75$) and $+0.32$ ($\alpha=0.85$), and *reverses* to -0.36 at $\alpha = 0.95$. The marginal 0.993 is α -driven confounding. (This stratification was demanded by the project’s adversarial reviewer; the marginal number alone had survived our own first write-up.)

4.2 Finding 2: the wedge, and criticality escape

The (λ, α) surface of $\rho(SB)$ shows two regimes (Fig. 1). On potential games, ρ is *non-monotone* in λ : it rises to ≈ 0.5 near $\lambda \approx 1.7$ and then collapses to zero, because the high- λ principal branch concentrates on a strict equilibrium and $C \rightarrow 0$ exponentially — $S = \lambda C \rightarrow 0$ despite the λ factor. Potential games *escape* criticality through equilibrium concentration. At $\alpha = 1$, mixed equilibria keep C bounded away from zero and ρ grows without bound. In between, a supercritical wedge opens: the median game first crosses $\rho = 1$ at $\alpha = 0.5$ ($\lambda \geq 8.5$); a 0.2 fraction of individual games crosses already at $\alpha = 0.4$; the frontier descends to $\lambda_c \approx 3$ by $\alpha = 0.8$. Inside the wedge the resolvent is near-singular and $\mathcal{R}$ magnitudes are direction-only — the surface artifact carries this caveat cell-by-cell.

4.3 Finding 3: the response and dissipation layers are distinct (a refuted repair, reported plainly)

To explain Finding 1 we hypothesised that $\mathcal{R}$ ’s denominator $\|\chi^{eq} + \chi^{eq\top}\|_F$ underflows at high α (the ratio measuring the residual symmetric sliver), and that the *numerator* $\|\chi^{eq} - \chi^{eq\top}\|_F$ would remain a valid circulation gauge. We tested both on the identical seeded games (paired design). Half held: $\rho(\mathcal{R}, 1/\text{den} \mid \alpha)$ climbs monotonically to 0.993 at $\alpha = 0.95$ — at high α , $\mathcal{R}$ ’s within-level variation *is* the shrinking symmetric response. But the repair failed: the numerator also decouples, $\rho(\text{num}, \sigma_{EP} \mid \alpha=0.95) = -0.37$ (95% bootstrap CI excluding zero), against $+0.77$ to $+0.86$ for $\alpha \leq 0.65$. No renormalisation of the response matrix recovers the dissipation ordering in the near-harmonic regime.

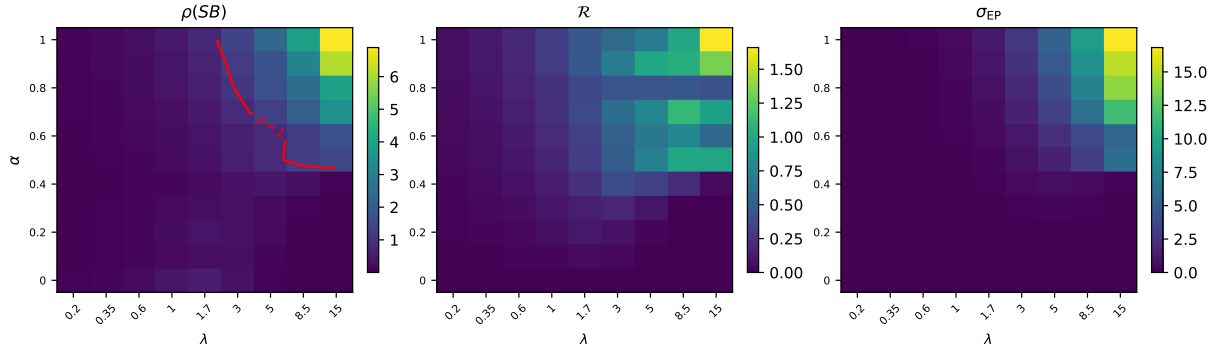


Figure 1: The (λ, α) surface: median $\rho(SB)$, $\mathcal{R}$, and σ_{EP} over seeded families (5 games/cell; α increases upward, λ rightward). The criticality escape at $\alpha = 0$ and the supercritical wedge are visible in the left panel. Regenerates from `experiments/phase_map.py`, seed 20260808.

Our reading — argued, not proved — is that χ^{eq} is a local derivative at the QRE point (a marginal-strategy object) while σ_{EP} is a functional of the joint-profile stationary flux; nothing identifies one with the other except in potential games, where both vanish. The practical scope table this yields: $\mathcal{R}$ answers “*is this system potential?*” exactly, at every λ and every α ; σ_{EP} answers “*how hard is it circulating?*”; $\mathcal{R}$ levels are comparable across systems only for $\alpha \lesssim 0.65$ (measured at $\lambda = 1.2$, 3×3 , two players; the crossover’s λ - and size-dependence is open). The refuted hypothesis, its post-hoc origin, and the sequence of events are recorded in the project’s findings log; the gate protecting this result was specified after measurement as a regression contract, and says so on its face.

5 Software and reproducibility

`strataq` is a uv-workspace monorepo (github.com/SharathSPhD/sage) in JAX with float64 enforced at import — the susceptibility and dissipation computations are ill-conditioned near criticality and float32 is not acceptable there. Architecture in brief: three mathematical *engines* (finite strategic form; population games; Bayesian, deferred by recorded decision), with solvers, implicit differentiation and the dynamics/entropy machinery shared in an engine-agnostic core. Domains — congestion, Blotto, pricing, electricity — are *plugins*: exactly five declared objects (payoff oracle, action-grid builder, conjugate-field spec, loader, learn page) and zero core changes, enforced by a commit hook and CI. The conjugate-field declaration is load-bearing: a domain that cannot name an observable payoff perturbation (e.g. penalty kicks) declares `NONE`, and the response instruments *refuse* to run for it rather than returning a meaningless number.

Reproducibility is contractual, not aspirational. A work unit closes only when its *gate* — code (tests, strict types, lint, coverage, no stubs), domain (the claim demonstrably true, with machine-readable artifacts), statistical (effect sizes with intervals, justified n , recorded seeds), documentation, and adversarial (a red-team review that sees the artifact and claim but never the implementation rationale) — is green, and every domain artifact must regenerate from its recorded seed via `make reproduce`. Anti-gaming rules are mechanical: deleting or `xfail`-ing a failing test blocks the commit; weakening a domain gate requires a recorded architecture decision; gate regressions fail CI. The adversarial layer earned its keep: of the roughly twenty objections raised across the units behind this paper, three were blocking defects in our own

enforcement tooling, one exposed the marginal–conditional trap of Finding 1, and one forced the epistemic disclosures around Finding 3.

Performance notes for users: solvers are JIT-compiled (the 2,000-game α sweep with dissipation readings completes in ~ 20 s on CPU); the separable Hodge transform is near-linear in tensor size; the dense dynamics path is exact and intended for small games, with trajectory estimators (KLD, TUR bounds) planned for scale. `pygambit` is an optional oracle dependency, never a runtime requirement.

6 Related work and honest positioning

What is known, and cited as such. The entropy-regularised characterisation of logit response, the CGF structure of the log-partition, Gibbs stationarity of logit dynamics in potential games, and the symmetry-characterisation of potentiality are textbook [1, 5, 4, 7, 8, 14]. The intrinsic potential/harmonic decomposition is Candogan et al. [6]. An earlier draft of the research programme claimed the symmetry characterisation as new; the project’s own sanity check caught and downgraded it — that correction is part of the claim ledger’s public history, and we mention it as evidence the tier system binds us, not only others.

Nearest live work. Legacci et al. [9] prove, in Shahshahani geometry, that harmonic games are incompressible under continuous-time exponential weights — volume preservation, constants of motion, Poincaré recurrence — and leave discrete-time behaviour and mixed games open. Our programme sits at a different layer: discrete logit QRE, the observable equilibrium-response matrix, and non-equilibrium observables (currents, dissipation), with calibration on real data. Their precision fix (potentiality of the *effective* game; Lemma C.2) is baked into our decomposition path. We found no prior statement of the reciprocity transfer (Result 1) or of response-matrix asymmetry as an observable proxy for harmonic content; the sweep and its scope are recorded in the repository’s literature file.

What this paper does not claim. $\mathcal{R}$ is not a universal circulation gauge (Finding 3 refutes exactly that); λ estimated from data absorbs unmodelled heterogeneity, and low λ is not proof of irrationality [3]; the Sioux Falls exercise is calibration on computed equilibria, not an empirical claim about drivers; and the thermodynamic language is exact only where the identities are exact — for non-potential games what survives is the NESS framing, which is where the interesting readings live. Empirical contact (scanner panels, electricity offer curves, λ estimator families and conduct testing) is the subject of the programme’s later papers; the closest empirical precedent for the dissipation angle is Xu and Wang [16], who identify an Edgeworth cycle via nonzero entropy production in experimental market data.

A Every number’s provenance

Every quantitative statement in this paper maps to a machine-readable artifact in `benchmarks/results/` (JSON, schema `strataq_bench.BenchmarkResult`: metrics, effect sizes with intervals, seed, config reference, library version, timestamp). All regenerate with `make reproduce` (seed 20260808 unless stated). The gate records in `gates/*.yaml` carry, for each unit, the claim, the section-by-section pass state, and the adversarial reviewer’s objections with their dispositions — including the two rounds on Finding 3’s unit, whose gate discloses on its face that its thresholds were specified post-measurement as a regression contract.

Statement in text	Artifact
Calibration table (Table 1)	the ten artifacts named in its rows
$\rho(\mathcal{R}, \alpha) = 0.982$, $n=2000$	<code>reciprocity_alpha_sweep.json</code>
$\rho(\sigma_{\text{EP}}, \alpha) = 0.990$; $\rho(\sigma_{\text{EP}}, \mathcal{R}) = 0.993$; within-level reversal	<code>chain_comovement.json</code>
Phase surface; wedge onset; criticality escape	<code>phase_map.json</code> , <code>phase_map_surface.json</code>
Finding 3 (H2 confirmed, H1 refuted, CIs)	<code>decoupling_mechanism.json</code>
Solver agreements (damped/mirror/Gambit/implicit/branch)	<code>solver_cross_agreement.json</code> , <code>gambit_agreement.json</code> , <code>implicit_chi_agreement.json</code> , <code>branch_agreement.json</code>
Sioux Falls calibration; UE diagnostic honesty note	<code>sioux_falls_calibration.json</code>
Engine-2 identities (Beckmann gradient, KKT, DF , toll- χ)	<code>population_identities.json</code>
Blotto readings incl. the degenerate null	<code>blotto_readings.json</code>

The findings log (`memory/findings.md`, entries F-0001–F-0007) records each anomaly with the config that produced it, its chase status, and its resolution — including the refuted hypothesis of Finding 3 and the provenance of its post-hoc origin. The claim ledger (`memory/claims.md`) carries every scientific claim’s confidence tier and history, including downgrades.

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